

Function Notation Is Not Multiplication

Answer Key

Algebra 1

Quick Answers

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|---|---|
| 1. (a) $f(3) = 5$, — | 6. (a) $t = 9$ min, — |
| 2. (a) $g(5) = 15$, — | 7. (a) $f(0) = -12$, (b) x when the output is 0 = 3, — |
| 3. (a) $f(5) = 26$, (a) $f(2) + f(3) = 15$, — | 8. (a) $f(3) = 15$, (a) $f(6) = 48$, — |
| 4. (a) $x = 4$, — | |
| 5. (a) cost of 4 shirts = 73, — | |
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What is verified

0 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSF-IF.A – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

5. *If they got 48: used only the per-shirt rate and dropped the fixed setup fee*

1. $f(x) = 4x - 7$; a student wrote $f(3) = f \cdot 3 = 3f$. (a) Evaluate $f(3)$. (b) Say what $f(3)$ instructs you to do.

Solution (a): The parentheses hold the *input*, so replace every x in the rule by 3:
 $4(3) - 7 = 12 - 7$. $f(3) = 5$

(b) — instructor-judged. A correct sentence says $f(3)$ is an instruction to substitute 3 for x in the rule and evaluate, not a product of a quantity f with 3. Full credit for naming substitution as the operation; partial credit for saying only that f is the name of the function.

2. $g(x) = x^2 - 2x$; a student wrote $g(5) = 25g - 10g = 15g$. (a) Evaluate $g(5)$. (b) Name the error.

Solution (a): Substitute 5 for x — the letter g does not enter the arithmetic at all:
 $5^2 - 2(5) = 25 - 10$. $g(5) = 15$

(b) — instructor-judged. A correct answer says the student kept the function's name as if it were a factor, writing a g beside every term, instead of replacing the input variable x by 5. The letter naming the function never enters the arithmetic. Full credit for identifying the stray function name as the error; no credit for an answer that only recomputes the value.

3. $f(x) = x^2 + 1$; a student claims $f(2 + 3) = f(2) + f(3)$. (a) Compute both sides. (b) Explain why the claim fails.

Solution (a): Left side: the input is the whole sum, $2 + 3 = 5$, so $f(5) = 5^2 + 1 = 26$. Right side: $f(2) = 2^2 + 1 = 5$ and $f(3) = 3^2 + 1 = 10$, so the sum is 15.

$$f(5) = 26 \quad \text{and} \quad f(2) + f(3) = 15$$

(b) — instructor-judged. A correct answer says the parentheses in $f(2 + 3)$ hold the input, so the whole sum is substituted once and squared, while $f(2) + f(3)$ squares each number separately and adds the constant 1 twice. Splitting across a plus sign is a distributive move; substitution is not multiplication, so it is not allowed. Full credit for tying the failure to the parentheses holding an input rather than a factor.

4. $f(x) = 6x - 4$; a student solved $f(x) = 20$ as $f \cdot x = 20$, $x = \frac{10}{3}$. (a) Solve correctly. (b) Name the error.

Solution (a): $f(x) = 20$ asks which input makes the rule output 20, so solve $6x - 4 = 20$. Add 4: $6x = 24$. Divide by 6: $x = 4$. Check: $6(4) - 4 = 20$ ✓

$$x = 4$$

Common wrong answer: $x = \frac{10}{3}$ comes from reading the notation as a product and dividing 20 by 6 without first undoing the subtraction.

(b) — instructor-judged. A correct answer says the student treated the notation as a product of two quantities and divided, when the equation actually asks which input makes the rule output 20, so the rule must be undone term by term. Full credit for naming the false product step; partial credit for showing the correct solution with no diagnosis.

5. Order cost $C(n) = 12n + 25$ dollars; a student wrote $C \cdot 4 = 4C$. (a) Find $C(4)$ in dollars. (b) Explain the meaning of the 25.

Solution (a): Substitute $n = 4$: $12(4) + 25 = 48 + 25$.

$$C(4) = 73$$

Common wrong answer: 48 comes from using only the per-shirt rate and dropping the fixed setup fee.

(b) — instructor-judged. A correct answer states that 25 is the one-time setup charge in dollars, paid whatever the order size, and that the notation names the cost of an order of 4 shirts rather than four copies of some number. Full credit for both halves; partial credit for the meaning of the setup charge alone.

6. $A(t) = 45 - 5t$ litres after t minutes; a student read $A(t) = 0$ as “ A times t is 0”. (a) Solve $A(t) = 0$ in minutes. (b) Name the error.

Solution (a): $A(t) = 0$ asks for the time when nothing is left, so solve $45 - 5t = 0$. Then $5t = 45$ and $t = 9$. Check: $45 - 5(9) = 0$ ✓

$$t = 9 \text{ min}$$

Common wrong answer: $t = 0$ comes from reading the notation as a product set equal to zero and concluding that one factor must vanish.

(b) — **instructor-judged.** A correct answer says that a product being zero forces a factor to be zero, but this expression is not a product of two quantities: it is the amount of water left after t minutes, so setting it to zero asks when the tank empties. Full credit for rejecting the zero-product move by pointing out that there is no product.

7. $f(x) = 4x - 12$. (a) Find $f(0)$. (b) Solve $f(x) = 0$. (c) Explain the difference.

Solution (a): Substitute 0 for the input: $4(0) - 12 = -12$.

$$f(0) = -12$$

Solution (b): Here the *output* is fixed at 0 and the input is unknown: $4x - 12 = 0$, so $4x = 12$ and $x = 3$.

$$x = 3$$

(c) — **instructor-judged.** A correct sentence says the first question fixes the input at zero and asks for the output, while the second fixes the output at zero and asks for the input; one is an evaluation, the other is an equation to solve. Full credit for naming which quantity is given in each.

8. $f(x) = x^2 + 2x$; a student claims $f(6) = 2f(3)$. (a) Compute $f(3)$ and $f(6)$. (b) Show the claim is false.

Solution (a): $f(3) = 3^2 + 2(3) = 9 + 6 = 15$ and $f(6) = 6^2 + 2(6) = 36 + 12 = 48$.

$$f(3) = 15 \quad \text{and} \quad f(6) = 48$$

(b) — **instructor-judged.** A correct answer computes twice the value at input 3, gets 30, compares it with 48, and says the two disagree, so doubling the input does not double the output. It then states that the notation instructs substitution of 6 into the rule, not multiplication of a quantity by 6. Full credit for the numerical comparison plus the statement about substitution.